Project Design Phase

1. Introduction

The **Project Design Phase** focuses on designing the structure and architecture of the system before development begins.
This phase defines **how the system components interact, how data flows through the system, and how the database tables are organized**.

The design ensures that the system can efficiently **import external data and automatically map relationships between tables using Import Sets and Dot Walking**.

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2. System Architecture Design

The system architecture consists of several components that work together to automate data import and mapping.

Components

Component	Description
Data Source	External CSV or Excel files
Import Set Loader	Uploads and processes external data
Import Set Table	Temporary storage for imported data
Transform Map	Maps source fields to target fields
Dot Walking	Connects related tables
Target Tables	Stores final processed data

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3. High-Level System Design

External Data File
|
▼
Import Set Loader
|
▼
Import Set Table
|
▼
Transform Map
|
▼
Dot Walking (Relationship Mapping)
|
▼

Target Tables
(User / Team / Department)

This design ensures **smooth data processing from the source to the final database tables**.

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4. Database Design

The system uses multiple tables to store and manage the data.

Main Tables

Table Name	Purpose
Import Set Table	Temporary storage for imported data
User Table	Stores user information
Team Table	Stores team details
Department Table	Stores department information

Example Table Relationship

```
User Table
  |
  ├── team_id
  |
  ▼
Team Table
  |
  ├── department_id
  |
  ▼
Department Table
```

Using **Dot Walking**, the system can access related data like:

user.team.department.name

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5. Module Design

1. Data Import Module

Handles uploading and reading external files.

2. Data Transformation Module

Maps fields between Import Set tables and target tables.

3. Relationship Mapping Module

Uses **dot walking** to link related tables automatically.

4. Data Validation Module

Checks for errors, missing values, or duplicate records.

5. Data Storage Module

Stores validated data in the final database tables.

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6. User Interface Design

The system interface allows administrators to:

- Upload CSV/Excel files
- View imported records
- Monitor data transformation
- Confirm successful data import

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7. Security Design

Security features include:

- User authentication
- Role-based access control
- Data validation before database insertion

These features ensure that **only authorized users can perform data import operations**.

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8. Expected Design Outcome

The system design ensures:

- Efficient data import from external files
- Automatic relationship mapping
- Reduced manual work
- Improved data accuracy